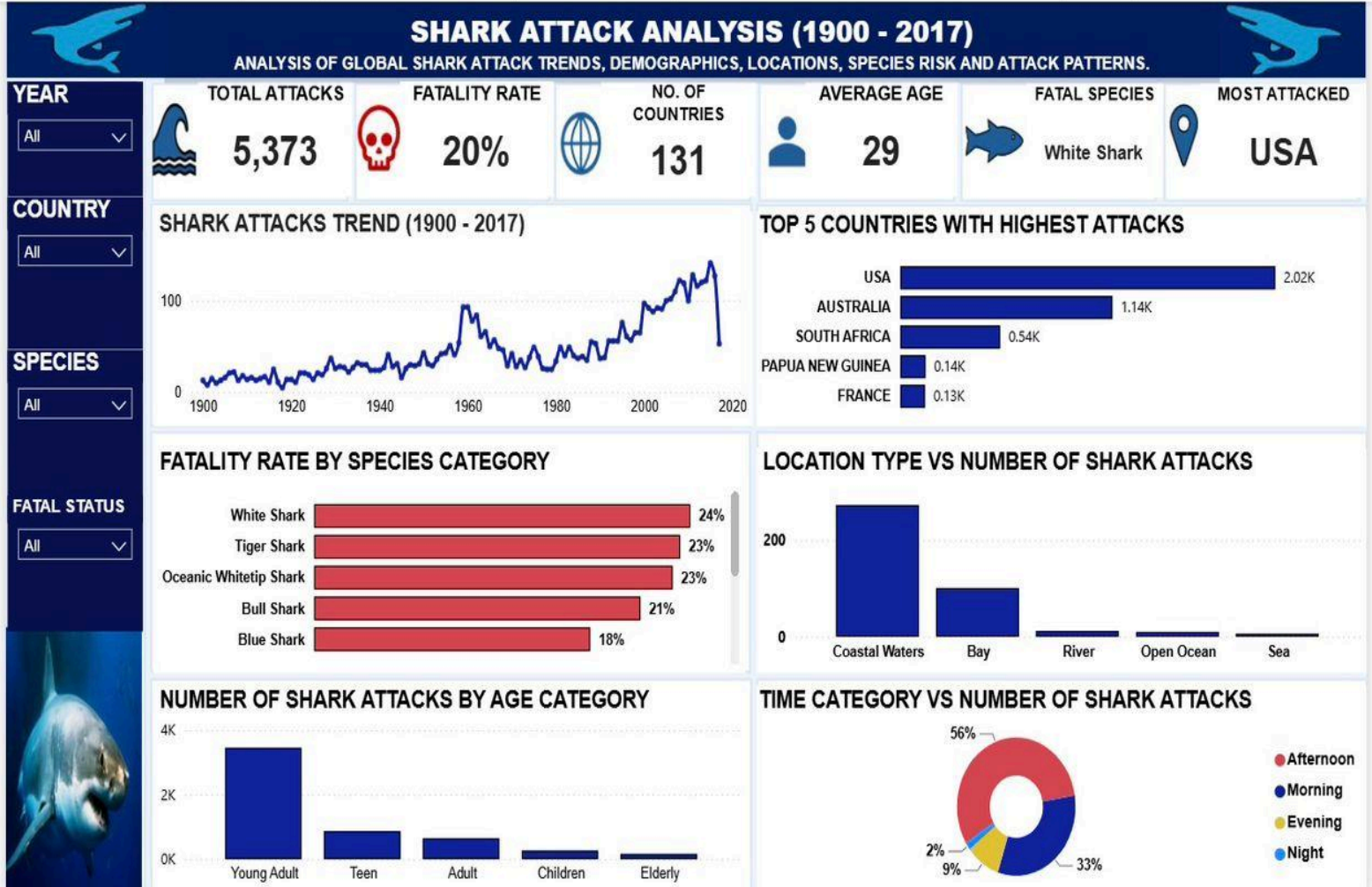


SHARK ATTACK ANALYSIS DASHBOARD



Introduction

This report presents a comprehensive analysis of global shark attack data. The primary objective is to identify geographic hotspots, evaluate environmental factors influencing incident frequency, and provide actionable insights to support safety awareness and management decision-making. The analysis was performed using Power BI, where the dataset underwent thorough cleaning, transformation, modeling, and visualization to create an interactive dashboard highlighting critical trends.

Data Overview

The dataset contains detailed information regarding historical shark attacks, including:

- Date of incident
- Time of incident
- Geographic location (Country and Area)
- Environmental context (Location Type)
- Activity Types
- Incident outcomes

This dataset provides a foundation for understanding the frequency and distribution of shark-human interactions.

Data Cleaning and Preparation

To ensure the accuracy and reliability of the analysis, the dataset was processed in Power BI Power Query. Key preparation steps included:

Handling Outliers: To ensure the analysis focused on relevant and reliable contemporary data, we constrained the dataset by filtering for incidents occurring between 1900 and 2017. This approach effectively excluded historical outliers and data points outside of the desired temporal scope, ensuring that the results reflect modern shark-human interaction patterns.

Handling Missing Values: To ensure statistical consistency, missing age entries were populated using gender-stratified mean imputation, assigning the average age of the corresponding male or female demographic to the affected records. For other categorical fields, missing entries were assigned to an "Unknown" or "Other" category to prevent the loss of significant data points.

Data Standardization: Geographic data, specifically the Area column, was cleaned to standardize entries into consistent State/Province/Region categories.

Age Standardization: The Age numerical data was grouped into the following standardized categories:

- **Children:** 0–12 years
- **Teen:** 13–19 years
- **Young Adult:** 20–35 years
- **Adult:** 36–59 years
- **Elderly:** 60+ years

- **Activity Standardization:** Activities were standardized into the following 17 categories:

- Boarding
- Boating
- Collecting
- Diving
- Filming
- Fishing
- Jumping
- Murder
- Other
- Rescue
- Sea Disaster
- Shark Interaction
- Standing
- Surfing
- Swimming
- Unknown
- Walking

Injury Standardization:

- **Injury Category:** Standardized into 14 types:

- Abrasion
- Bite
- Bruise
- Fatal
- Fracture
- Laceration
- Minor Injury
- Multiple Injuries
- No Injury
- No Shark Injury
- Puncture
- Severed/Amputation
- Unknown
- Wound

● **Body Part Injury Standardization:** Standardized into 11 categories:

- Arm Injuries
- Fatal
- Foot Injuries
- Hand Injuries
- Head Injuries
- Leg Injuries
- Multiple Body Injuries
- No Body Injury
- Thigh Injuries
- Torso Injuries
- Unknown Injuries.

Time Standardization: Incident times were standardized into four distinct temporal categories:

- Morning
- Afternoon
- Evening
- Night

Species Standardization: Shark species were mapped into the following 20 categories:

- Blacktip Shark
- Blue Shark
- Bronze Whaler Shark
- Bull Shark
- Dusky Shark
- Grey Nurse Shark
- Hammerhead Shark
- Lemon Shark
- Mako Shark
- Multiple Species
- No Shark Involvement
- Nurse Shark
- Oceanic Whitetip Shark
- Porbeagle Shark
- Reef Shark
- Sevengill Shark
- Tiger Shark
- Unidentified Species
- White Shark
- Wobbegong Shark.

Location Type Standardization: Incident environments were standardized into the following 8 categories:

- Bay
- Coastal Waters
- Gulf
- Lake
- Open Ocean
- River
- Sea
- Unknown.

Data Types: Appropriate data types were assigned to all columns to ensure correct mathematical calculations and date-based filtering.

Data Modelling

Following data cleaning and transformation, a star schema data model was developed in Power BI to improve analytical efficiency and support dashboard reporting. The model was designed to separate descriptive attributes from incident level records, thereby reducing redundancy and improving query performance.

Two dimension tables and one fact table were created:

Fact Table

The Fact_Shark Attacks table served as the central table containing shark attack incidents. Each record represented a unique shark attack case and included key identifiers used to establish relationships with the dimension tables.

Dimension Tables

Dim Victim

The Dim Victim table contained demographic information relating to shark attack victims. Attributes included:

- Case Number
- Name
- Age
- Age Group
- Gender

This table enabled demographic analysis of shark attack incidents.

Dim Injury

The Dim Injury table contained information related to attack outcomes and incident characteristics. Attributes included:

- Case Number
- Year
- Type
- Fatality Status (Y/N)
- Species Category
- Country
- Continent
- Area
- Location Type
- Time Category
- Injury Category
- Body Part Injury Category

This table supported analysis of attack severity, species involvement, geographical distribution, and temporal patterns.

Relationships

Relationships were established between the fact table and dimension tables using the Case Number field. This allowed filters and slicers to propagate efficiently throughout the model while maintaining data integrity.

The resulting model provided a structured framework for performing descriptive and comparative analyses of global shark attack incidents occurring between 1990 and 2017.

Measures and Calculations

Several DAX measures were created to support key performance indicators (KPIs) and dashboard visualizations.

Total Shark Attacks

This measure calculates the total number of shark attack incidents recorded within the study period.

Total Fatalities

This measure calculates the total number of fatal shark attack incidents.

Fatality Rate

This measure calculates the proportion of shark attacks that resulted in fatalities. The measure was formatted as a percentage for reporting purposes.

Total Countries Affected

This measure calculates the number of countries that experienced shark attacks while excluding records classified as "Unknown".

Average Victim Age

This measure calculates the average age of shark attack victims.

Most Fatal Species

This measure identifies the shark species responsible for the highest number of fatal attacks while excluding unidentified species and incidents involving no shark participation.

The measure was developed using DAX ranking logic to determine the species associated with the greatest number of fatalities.

Most Attacked Country

This measure identifies the country with the highest number of recorded shark attacks, excluding unknown locations.

The measure uses ranking functions to determine the country with the largest attack count.

Dashboard Overview

An interactive Power BI dashboard was developed to analyze and visualize global shark attack incidents recorded between 1900 and 2017. The dashboard transforms cleaned and modelled data into meaningful insights through the use of key performance indicators (KPIs), charts, and interactive filters. The objective of the dashboard is to provide a comprehensive understanding of shark attack patterns by examining temporal trends, demographic characteristics, geographical distribution, species related risks, and behavioral factors associated with shark attacks.

The dashboard was designed to support data driven decision making by enabling users to explore attack incidents from multiple perspectives and identify significant patterns within the data.

Dashboard Objectives

The dashboard was developed to:

- Analyze trends in shark attacks between 1990 and 2017.
- Identify countries with the highest number of shark attacks.
- Examine the fatality rates associated with different shark species.
- Explore the demographic characteristics of shark attack victims.
- Investigate activities most commonly associated with shark attacks.
- Identify patterns in attack occurrence across different times of the day.

Key Performance Indicators (KPIs)

The dashboard includes six KPI cards that provide a high level summary of the shark attack dataset.

Total Shark Attacks

Displays the total number of recorded shark attack incidents between 1990 and 2017.

Fatality Rate

Shows the percentage of shark attacks that resulted in fatalities.

Total Countries Affected

Indicates the number of countries that recorded shark attack incidents, excluding unknown locations.

Average Victim Age

Displays the average age of shark attack victims.

Most Fatal Species

Identifies the shark species associated with the highest number of fatal attacks, excluding unidentified species and incidents with no shark involvement.

Most Attacked Country

Displays the country with the highest number of recorded shark attack incidents.

Dashboard Visualizations

Attack Trends (1900–2017)

A line chart displaying the annual number of shark attacks over the study period. This visualization enables the identification of long term trends, fluctuations, and changes in attack frequency over time.

Attack Count by Age Group

A column chart illustrating the distribution of shark attacks across different victim age groups. This visualization highlights the age categories most frequently affected by shark attacks.

Species Category vs Fatality Rate

A bar chart comparing shark species categories based on their fatality rates. This visual helps identify species that pose the greatest risk of death when an attack occurs.

Top 5 Countries with Highest Attacks

A horizontal bar chart displaying the five countries with the highest number of recorded shark attack incidents. This visualization highlights geographical hotspots and areas with elevated shark attack activity.

Location Type vs Number of Shark Attacks

A bar chart showing the frequency of shark attacks across different location categories, such as Coastal Waters, Bay, River, Open Ocean and Sea. This visualization shows location categories associated with higher exposure to shark attacks.

Attacks by Time Categories

A chart illustrating the distribution of shark attacks across different periods of the day, including dawn, morning, afternoon, evening, night, and midnight. This visualization helps identify temporal patterns and peak periods for shark attack occurrences.

Interactive Features

The dashboard incorporates interactive filtering and cross filtering capabilities that enable users to explore the data dynamically. Users can filter results based on attributes such as country, shark species category, Year and fatality status. These interactive features enhance the analytical value of the dashboard by allowing users to investigate specific trends and relationships within the dataset.

Dashboard Value

The dashboard provides a centralized analytical platform for understanding shark attack incidents from demographic, geographical, temporal, behavioral, and species related perspectives. By integrating KPI summaries with interactive visualizations, the dashboard enables users to quickly identify trends, assess risks, and gain deeper insights into the factors associated with shark attack occurrences and outcomes.

Geographic Analysis

- The geographic analysis examined the distribution of shark attacks across countries between 1900 and 2017.
- Key findings include:
- The majority of shark attacks were concentrated in a small number of countries.
- The United States recorded the highest number of shark attacks overall.
- Australia and South Africa were among the countries with the highest attack frequencies.
- Several countries recorded relatively few shark attack incidents compared to the leading countries.
- These findings indicate that shark attacks are not evenly distributed globally but are concentrated in specific coastal regions. The high number of attacks in countries such as the United States and Australia may be attributed to extensive coastlines, high participation in water related activities, and greater reporting of shark attack incidents.

Demographic Analysis

The demographic analysis focused on the age distribution of shark attack victims.

Key findings include:

- Shark attacks were most common among young and middle aged individuals.
- Certain age groups recorded substantially higher attack counts than others.
- The average victim age was approximately [Insert Average Age] years.
- Older age groups experienced comparatively fewer shark attacks.
- These findings suggest that individuals within active age groups are more likely to engage in water based recreational activities, increasing their exposure to shark encounters. The results indicate that age may be an important factor influencing shark attack risk.

Species Analysis

The species analysis evaluated shark attack outcomes across different shark species categories.

Key findings include:

- • White sharks were responsible for the highest number of fatal attacks.
- • Tiger sharks recorded one of the highest fatality rates among identified species.
- • Fatality rates varied considerably across shark species categories.
- • Certain species were associated with frequent attacks but relatively lower fatality rates.
- These findings indicate that attack frequency and fatality risk differ significantly among shark species. While some species are involved in a greater number of attacks, others pose a higher mortality risk when attacks occur.

Activity Analysis

The activity analysis examined the location victims were at the time of the attack.

Key findings include:

- Coastal waters recorded the highest number of shark attacks.
- Bays also recorded quite high number of shark attacks
- Rivers, Open ocean and Sea recorded very low numbers of shark attacks.
- These findings suggest that Coastal Waters are associated with greater exposure to shark encounters. Individuals participating in activities that place them in coastal waters for extended periods may face a higher likelihood of shark interactions.

Time Based Analysis

The time based analysis examined shark attack occurrences across different periods of the day.

Key findings include:

- The majority of shark attacks occurred during the afternoon.
- Morning periods also recorded a significant number of attacks.
- Evening attacks were less frequent.
- Night time attacks represented the smallest proportion of incidents.

- These findings indicate that shark attacks are more likely to occur during periods when human water activity is highest. The concentration of attacks during daylight hours may reflect increased participation in swimming, surfing, and other recreational activities.

- **Fatality Analysis**

- The fatality analysis assessed the severity of shark attacks and their outcomes.

Key findings include:

- The overall fatality rate was approximately (20%)
- Most shark attacks did not result in fatalities.
- Fatal outcomes were strongly influenced by shark species involved in the attack.
- Certain species exhibited significantly higher fatality rates than others.
- These findings suggest that although shark attacks attract significant public attention, fatal incidents represent a relatively small proportion of total attacks. However, attack outcomes can vary substantially depending on the species involved and the circumstances surrounding the incident.

Trend Analysis

The trend analysis evaluated changes in shark attack frequency between 1900 and 2017.

Key findings include:

- Shark attack incidents generally increased over the study period.
- Several years recorded notable spikes in attack activity.
- The highest attack counts occurred during the later years of the study period.
- Despite annual fluctuations, the overall trend remained upward.
- These findings suggest that shark attack reporting and occurrence increased between 1900 and 2017. Factors such as increased coastal recreation, population growth, tourism, and improved reporting systems may have contributed to the observed trend.

Recommendations

- **Implement Targeted Public Awareness Campaigns:** Focus safety messaging on **Coastal Waters** and **Bay** environments, particularly during **Afternoon** hours, as historical data indicates these settings and times have a higher frequency of interactions.
- **Develop Activity-Specific Safety Protocols:** Establish and promote safety guidelines tailored to high-susceptibility activities, specifically **Surfing** and **Swimming**, to minimize risk during these common behaviors.
- **Enhance Demographic-Specific Educational Outreach:** Create and distribute educational materials aimed at the **Teen** and **Young Adult** age cohorts, as these groups demonstrate a higher propensity for shark interactions.
- **Support Evidence-Based Local Policy:** Utilize species-specific insights to assist local authorities in regions with higher concentrations of **White Shark** or **Bull Shark** activity, allowing for more precise management and public warning systems.
- **Standardize Emergency Medical Training:** Prioritize emergency response training for **Foot** and **Leg** injuries, given their high frequency in the dataset, to improve the effectiveness of rapid, on-site medical intervention.
- **Data Maintenance:** Utilize the established geographic mapping table to standardize any future data entries, ensuring the dashboard remains accurate as new information is added.
- **Prioritize High-Risk Hotspots:** Allocate safety resources, surveillance, and public awareness campaigns to the high-incident countries and standardized areas identified in the dashboard.

Question 1:

Make a graph of the number of shark attacks annually over time since 1900. What trends do you see?

The annual shark attack trend analysis was conducted using a line chart showing the number of shark attacks recorded each year from 1900 to 2017.

Key findings include:

- Shark attacks generally increased over time.
- Relatively few attacks were recorded during the early 1900s.

- Attack frequencies increased substantially during the latter half of the twentieth century.
- The highest number of attacks occurred in the more recent years of the dataset.

The analysis suggests a long term upward trend in shark attack incidents. This increase may be attributed to population growth, increased participation in water related recreational activities, coastal tourism, and improved reporting of shark attacks over time.

Question 2:

Which countries report the most shark attacks? Within those countries, which areas and locations seem to be the most dangerous?

The geographic analysis examined shark attack distribution across countries, areas, and locations.

Key findings include:

- The United States recorded the highest number of shark attacks.
- Australia and South Africa were among the countries with the highest attack frequencies.
- A small number of countries accounted for the majority of reported shark attacks.
- Within these countries, certain coastal areas consistently recorded higher attack frequencies than others.

The findings indicate that shark attacks are concentrated in specific geographical regions rather than being evenly distributed worldwide. Coastal areas with high levels of water based recreational activities appear to experience greater shark attack risk.

Question 3:

What data cleaning and preparation steps were required before analysis?

The dataset contained numerous inconsistencies, missing values, and unstructured text fields that required extensive cleaning before meaningful analysis could be performed.

Key data preparation activities included:

- Standardizing country names.
- Cleaning and categorizing shark species descriptions into consistent species categories.

- Transforming injury descriptions into standardized injury categories and body part injury categories.
- Cleaning and grouping age values into age categories.
- Standardizing time values into Morning, Afternoon, Evening, and Night categories.
- Handling missing, unknown, and inconsistent values.
- Creating calculated columns and measures to support dashboard analysis.

These cleaning procedures improved data quality, consistency, and analytical reliability.

Question 4:

What body parts are most often injured?

Text analysis was performed on the Injury column to identify commonly injured body regions.

Key findings include:

- Injuries to the legs were the most common.
- Arm injuries were also frequently reported.
- Foot and hand injuries accounted for a significant proportion of attacks.
- Head and torso injuries occurred less frequently.

The results suggest that lower extremities are most vulnerable during shark attacks, likely because they are often the first body parts encountered by sharks during swimming and surfing activities.

Question 5:

Are shark attacks more common during certain parts of the day?

The Time column was cleaned and categorized into four periods:

- Morning
- Afternoon
- Evening

- Night

The analysis revealed that:

- The majority of shark attacks occurred during the afternoon.
- Morning recorded the second highest number of attacks.
- Evening attacks occurred less frequently.
- Night recorded the lowest number of shark attacks.

These findings suggest that shark attacks are more common during daylight hours when human participation in water based recreational activities is highest.

Question 6:

Which species of shark are attacking most often?

The species data contained numerous naming inconsistencies and was cleaned into standardized species categories.

Key findings include:

- White sharks were among the most frequently reported species.
- Tiger sharks and bull sharks were also responsible for a large number of attacks.
- A considerable number of incidents involved unidentified shark species.
- Attack frequency varied significantly across species categories.

The findings indicate that a small number of shark species account for the majority of recorded attacks. White sharks, tiger sharks, and bull sharks were consistently among the most commonly involved species throughout the dataset.

Overall Conclusion

The analysis revealed that shark attacks have generally increased over time and are concentrated within specific countries and coastal regions. Most attacks occur during daylight hours and are associated with recreational water activities. Leg injuries were the most common outcome, while white sharks, tiger sharks, and bull sharks were the species most frequently involved in attacks. The study demonstrates the importance of data cleaning and standardization when working with real world datasets and highlights how business intelligence tools can be used to uncover meaningful patterns within complex data.